

CLASS XII · LECTURE 9

Visual Basic: Variables, Decisions and Loops

Option II · Two periods · Dim, Variant, Select Case, For...Next

By the end of this lecture you can

- Declare a variable with Dim and explain Variant.
- Write Select Case.
- Write a For loop, including one that steps by 2.
- Convert a simple formula into VB.

Dim and Variant

Dim declares a variable. Dim marks As Integer creates an integer named marks. Other types you should name are Single and Double for fractional numbers, String for text, Boolean for True or False, Long for a large whole number, and Currency for money. A constant is declared with Const PASS As Integer = 40.

DEFINITION Variant

A Variant is a variable that can hold almost any kind of value: a number, a string or a date. Dim x As Variant, or simply Dim x, makes x a Variant. It is flexible and slower, and it hides mistakes. Use a specific type unless the question asks for Variant.

Val turns text into a number. A text box always gives you a string. Val(Text1.Text) is how you do arithmetic on what the user typed. Without Val, the + operator may join strings instead of adding. The & operator joins strings on purpose, as in a label that reads Total followed by the number.

If and Select Case

If marks >= 40 Then ... Else ... End If. For several exact values, Select Case is clearer than a long chain of If. The paper asks for its structure.

```
Private Sub Command1_Click()
    Dim grade As String
    grade = Text1.Text
    Select Case grade
        Case "A"
            Label1.Caption = "Excellent"
        Case "B"
            Label1.Caption = "Good"
        Case Else
            Label1.Caption = "Keep working"
    End Select
End Sub
```

Program 9.1 Select Case on a grade letter typed into Text1.

The structure is: Select Case expression, then one or more Case lines, an optional Case Else, and End Select. Case Is >= 80 is legal when you are testing a range. There is no break. Visual Basic does not fall through into the next case the way C does. Say that if you are a student who has also seen C, and do not write break inside Select Case.

For...Next

The counter starts at the first value and stops after the last. If Step is omitted, the counter increases by 1. Step 2 counts even numbers if you start at 0. Step -1 counts downwards. A Do While ... Loop repeats while a condition stays true. For a known count, use For.

```
Private Sub Command1_Click()
    Dim i As Integer
    For i = 0 To 50 Step 2
        Print i
    Next i
End Sub
```

Program 9.2 Even numbers from 0 to 50. Step 2 is the whole point of the program.

```
Private Sub Command1_Click()
    Dim i As Integer
    Dim n As Double
    Dim total As Double
    total = 0
    For i = 1 To 10
        n = Val(InputBox("Enter number " & i))
        total = total + n
    Next i
    Print "Sum = "; total
    Print "Average = "; total / 10
End Sub
```

Program 9.3 Sum and average of ten numbers. Val is required because InputBox returns text.

Formulas

The same translation rules as C apply, with VB's words. A square is $r * r$ or $r ^ 2$. A half is $0.5 * base * height$. Absolute value is $Abs(b * b - 4 * a * c)$. Pi can be written 3.1416. Integer division uses $\backslash$. Ordinary division uses $/$. You will usually want $/$.

Checkpoint

Try the questions before you read the model answers. Section B answers on the real paper should stay within about six or seven lines.

Multiple choice

1. Dim is used to:

- A) Declare a variable
- B) Close a form
- C) Draw a circle
- D) Send email

2. If Step is omitted in For...Next, the counter changes by:

- A) 0
- B) 1
- C) 10
- D) -1

Short answers

S1. How is a variable declared as Variant? Why might you avoid it?

S2. Write the structure of Select Case.

Model answers

Cover this section while you attempt the checkpoint. The wording is a full-mark style answer, not the only acceptable wording.

MCQ 1. A. Dim marks As Integer declares marks.

MCQ 2. B. The default step is 1.

S1. Dim x As Variant, or Dim x with no type, makes x a Variant. It can store a number, a string or a date. It is convenient and it is slow, and a mistaken text value will not be caught as a type error. A specific type such as Integer or Double is safer.

S2. Select Case expression, then Case value, then the statements for that value, further Case lines as needed, an optional Case Else, and End Select. Example: Select Case grade / Case "A" / Label1.Caption = "Excellent" / Case Else / Label1.Caption = "Other" / End Select. VB does not fall through to the next case.